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Golden Opportunity, or a New Twist on the Resource–Conflict Relationship: Links Between the Drug Trade and Illegal Gold Mining in Colombia

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Summary. — Resource wars face greater difficulties to end conflict, as well as greater probabilities of relapse. In part, this is due to the persistence of resource-fueled criminal networks developed under the auspices of armed conflict. In this paper we focus on the Colombian armed conflict, one of the longest-lasting conflicts in the world. Recent evidence suggests that gold mining in Colombia has been permeated by illegal organizations linked to the drug trade, driving armed conflict and criminality. This reveals that attention to drugs alone as a conflict resource in this particular case has overshadowed the degree to which legal resources and economic activities have been permeated by illegal organizations and interests. This paper provides a framework of the gold–drugs relationship, which reveals the existence of *resource portfolios*, or the parallel participation and exchangeability of resources in the provision of funding for illegal organizations. We argue that, in addition to the impact of each resource on armed conflict and criminality, illegal organizations develop abilities to extract benefits of different resources at once or interchangeably (a resource portfolio), which should be taken into account when analyzing the consequences of war on countries' social and economic institutions. In addition, political or reputational factors have been insufficiently considered in analyzing groups' decisions to engage in or abandon specific economic activities. We show that, along with expectations of revenue, resource portfolios may also respond to political conditions, as illegal organizations accustomed to deriving income from coercive practices such as kidnappings—until recently a widespread phenomenon in Colombia—have caused increasing international and domestic outrage followed by pressure to stop this brutal violation of Human Rights. Based on field research in gold mining Colombian regions—combining more than seventy semi-structured interviews with first-hand observation during field trips—and a careful review of press, non-governmental organizations' and official reports in local, regional, and national media, the paper provides a general framework of this complex relationship, paying specific attention to the evolution of the links and interchangeable nature of gold and drugs as conflict resources throughout the production phases of the gold extraction process. At a time when Colombia's ongoing peace process is likely to put an end to the armed confrontation between guerrilla groups and the Colombian state, our paper raises a warning sign for scholars and policymakers to consider the potential transformations of illicit markets and their role in shaping the prospects of durable peace.

Key words — resource wars, resource portfolio, armed conflict, gold mining, drug trade, Colombia

1. INTRODUCTION

Resource wars—armed conflicts in which resources are not only a source of funding, but in some cases have unleashed the confrontation (Le Billon, 2013)—are especially difficult to resolve and more likely to relapse than other types of conflict (Rustad & Binningsbø, 2012). This is due in part to the persistence of resource-fueled criminal networks developed under the auspices of armed conflict. In this paper we focus on the Colombian armed conflict, [1] one of the longest-lasting conflicts in the world. Since 2012, formal peace negotiations among the government and the leftist guerrilla FARC have sought to bring an end to the armed confrontation and provide non-repressive alternatives to the historical reliance of armed structures and civilian communities on the illegal production and distribution of coca leaves and paste. However, the shift away from illegal drugs—both as a policy focus and as an economic activity that sustains armed organizations—does not necessarily entail the weakening of criminal networks or an end to all criminal threats.

The difficulties associated with breaking the resource–conflict–crime connection will be discussed here by addressing what the former Commander of the Colombian Police, General Óscar Naranjo, depicted as a phenomenon "worse than the drug trade": illegal gold mining (*Caracol Radio,* March 23, 2012). Peaks in the international price of gold over the past decade were reflected in a fivefold increase of foreign investment in mining activities. Expectations of gold royalties'

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contributions to economic growth thus earned the mining sector unprecedented public policy attention (Ministerio de Minas y Energía, 2006). However, authorities and scholars also found that gold mining—especially illegal and informal operations, undertaken both by communities and illegal organizations—has been a source of revenue and strength for existing criminal networks in many Colombian regions at least since 2007 (Giraldo *et al.*, 2013; Giraldo & Muñoz, 2012; Idrobo, Mejía, & Tribín, 2013; Mejía, 2012; Rettberg, Norza, Ortiz-Riomalo, Romero, & Romero, 2013; Rettberg & Ortiz-Riomalo, 2013). This reveals that attention to drugs alone as a conflict resource has overshadowed the degree to which legal resources have been permeated by illegal organizations and interests (Rettberg, 2010, 2012; Rettberg, Leiteritz, & Nasi, 2011).

The links between armed conflict, crime, drugs, and gold in Colombia underscore one of the most long-lasting legacies of resource-fueled conflicts, namely the transformation and atrophy of social, political, and economic institutions by practices and incentives associated with the development of wartime illicit markets (García & Llinás, 2012; Rodrik, 2007; Rodrik, Subramanian, & Trebbi, 2004). These links also underscore the ongoing risks and challenges for peace and security in transitional countries such as Colombia over and beyond the specific impacts of the drug trade on political, economic, and social institutions.

In this paper we describe and analyze the links between gold mining, the drug trade, conflict, and crime in Colombia. Based on field research in gold mining Colombian regions—combining more than sixty semi-structured interviews with first-hand observation during field trips (details below)—and a careful review of press, non-governmental organizations' and official reports in local, regional, and national media, the paper provides an in-depth analysis of this complex relationship, paying specific attention to the evolution of the links between gold and drugs and their interchangeability as conflict resources throughout the production phases of the gold extraction process. We refer to the simultaneous participation in the extraction of multiple resources as funding sources for illegal organizations as a *criminal resource portfolio*.

The research presented here is relevant both for scholarly inquiry and for policy formulation. From a scholarly perspective, the links between resources and conflicts have been addressed by numerous authors (Arnson & Zartman, 2005; Collier & Hoeffler, 1998, 2004, 2005; Fearon, 2005; Keen, 1998). These authors have pointed out not only that financial viability—often derived from resource exploitation—is crucial for insurgent movements to prosper, but also that we need to distinguish among resources in terms of their contribution to armed conflict and crime. Factors such as resource lootability, obstruction, and legality (Ross, 2004), the geographical location of productive assets (Le Billon, 2001), and the type of institutions that govern the production or extraction of different resources have been included in efforts to trace when, why, and how resources become linked to conflict (Bannon & Collier, 2003; Berdal & Malone, 2000; Collier & Hoeffler, 2004; Elbadawi & Sambanis, 2002; Fearon, 2005; Keen, 1998; Ross, 2003).

The contribution of this paper to this literature resides in the effort to broaden the lens from the impact of individual resources to the links between different types of conflict-reinforcing resources, in this case gold and drugs. We find that illegal organizations develop capabilities to extract benefits from different resources either simultaneously or interchangeably—a dynamic that we refer to as the development of a resource portfolio. This dynamic, moreover, has an impact on countries' social and economic institutions in the context of war. In addition, we argue that political and reputational factors have been insufficiently considered in analyzing groups' decisions to engage in or abandon specific economic activities. The emphasis has been placed mainly on weak economic institutions or the absence of regulations facilitating criminal activity (Collier & Hoeffler, 2005). However, as will be shown here, in addition to revenue-related expectations, resource portfolios may also respond to political conditions, as illegal organizations accustomed to deriving income from coercive practices such as kidnappings—until recently a widespread phenomenon in Colombia (Ministerio de Defensa Nacional, 2011)—have faced increasing international and domestic pressure to comply with human rights and abandon such practices.[2] In a recent press interview, for example, one of the FARC's highest-ranking commanders said: "There was too much killing of informants. We could have behaved differently, capture them, call international organizations. Why kill them if all that was left were orphans? Kidnappings… we went too far. We made mistakes in exchange for money. We need to apologize for this. It's a fact" (Ruiz, 2015). In brief, resource portfolios are dynamic, responding not only to the logic of markets (international price, supply, and demand) but also to political motivations, such as the likelihood of coercion, and political and legitimacy costs associated with funding practices.

In addition, an extensive literature on peacebuilding has posited that the likelihood of conflict recurrence increases in transitional contexts when larger political economy networks making war sustainable are not addressed (Boyle, 2014; Keen, 1998; Le Billon, 2000; Rustad & Binningsbø, 2012). For example, some authors have referred to shadow economies, reared against a background of institutional weakness and economic opportunism by different armed actors (Nitzschke, 2003). Colombia's armed conflict has been related to the drug trade at least for the past two decades (Gaviria & Mejía, 2011; Ross, 2004). However, less attention has been paid to how illegal armed actors have systematically permeated economic and social institutions of production linked to other resources, thus posing a threat to the effective dismantling of incentive structures that link illegal actors to pervasive criminal activity.

Finally, this paper seeks to make a contribution to policymaking. Conflict-torn and post-conflict countries recurrently raise fears that lucrative resources will deepen domestic instability and crime. For example, oil discoveries in Uganda (Gelb & Majerowicz, 2011), a country with a long history of internal war, have led some to propose plain cash transfers to the population in order to stem incentives for corruption. Mozambique, also a country with a violent past, now faces the perspective of becoming one of the main global providers of coal and natural gas. This has put governments and international organizations on alert as to the possible ramifications into political instability and corruption (Biggs, 2012). Even beyond transitional contexts, gold mining has been linked to social unrest and economic instability, as documented by Hammond, Gond, de Thoisy, Forget, and DeDijn (2007) for the case of the Guiana Shield in the Northeastern Amazon region, by Theije and Heemskerk (2009) for the case of Suriname, and by Hilson and Potter (2005) for Ghana. Amid an ongoing yet difficult peace process, Colombia today faces a new opportunity for making a peaceful transition out of armed conflict. However, the larger economic and social processes supporting armed and criminal activity in Colombia need to be understood well so as to design comprehensive transitional strategies aimed at preventing spoiling and failure associated with old and new forms of resource-related violence. Policies need to focus on strengthening resource governance in order to delink natural resources from (armed) conflict, illegality, and crime.

The research process underlying this paper included fieldwork in five gold-mining regions (Antioquia, Caldas, Chocó, Nariño, Santander, and Tolima), during which we observed the conditions and local contexts of mining operations, and interviews with over seventy stakeholders, both at the national and regional levels. Our respondents included miners' organizations and mine owners and workers, state officials (including the Police, environmental authorities, and local authorities), academic and policy experts, and representatives of the large-scale mining industry. Table 1 presents the break-up of the interviewees by category.

The numbers of interviewees are not representative of any of these groups, but enabled us to collect a significant breadth of opinions and perspectives on subjects such as the conditions of mining operations, the perspectives of the mining sector, the way in which the armed conflict and the drug trade influenced mining operations and commercialization, and the strategies developed in response to local and national conditions. For security reasons, visits to the regions were announced beforehand and coordinated with local NGOs, associations, or international organizations present in the regions, so that all social and political actors were informed about the identity of the researchers and the nature of the research. The fact that the preparation for the visit to one of the mines included instructions as to how to behave in case we heard gun shots illustrates the suitability of this precaution. Field trips were complemented with a detailed review of official documents by state and sectoral actors, the existing scholarly literature, and press articles in order to triangulate information. Interview transcriptions were categorized by region and actor, as well as by different subjects. In the preparation of this document we paid specific attention to aspects related to the relationship between mining communities, armed conflict, and the drug trade.

The map below shows Colombian gold-mining regions in relation to illicit crop production. Circles depict the volume of gold production (in kg) in each of the selected regions (with Chocó and Antioquia ranking highest in national gold production). Map 1 illustrates regions in which mining titles have been allocated (light gray), coca crops are grown (medium gray), and where they overlap with gold mining operations (dark gray). The map visually conveys the frequency of this intersection and underscores the relevance of the analysis presented in the coming paragraphs.

The paper will first outline the conceptual underpinnings of research on resources, conflict, and peacebuilding. The next section will present a general overview of the relationship between gold, drugs, armed conflict, and crime in Colombia. Next, we will discuss the main findings and link them to the conceptual framework presented above. The conclusions provide suggestions for further research and some policy recommendations.

2. RESOURCES, CONFLICT, CRIME, AND PEACE

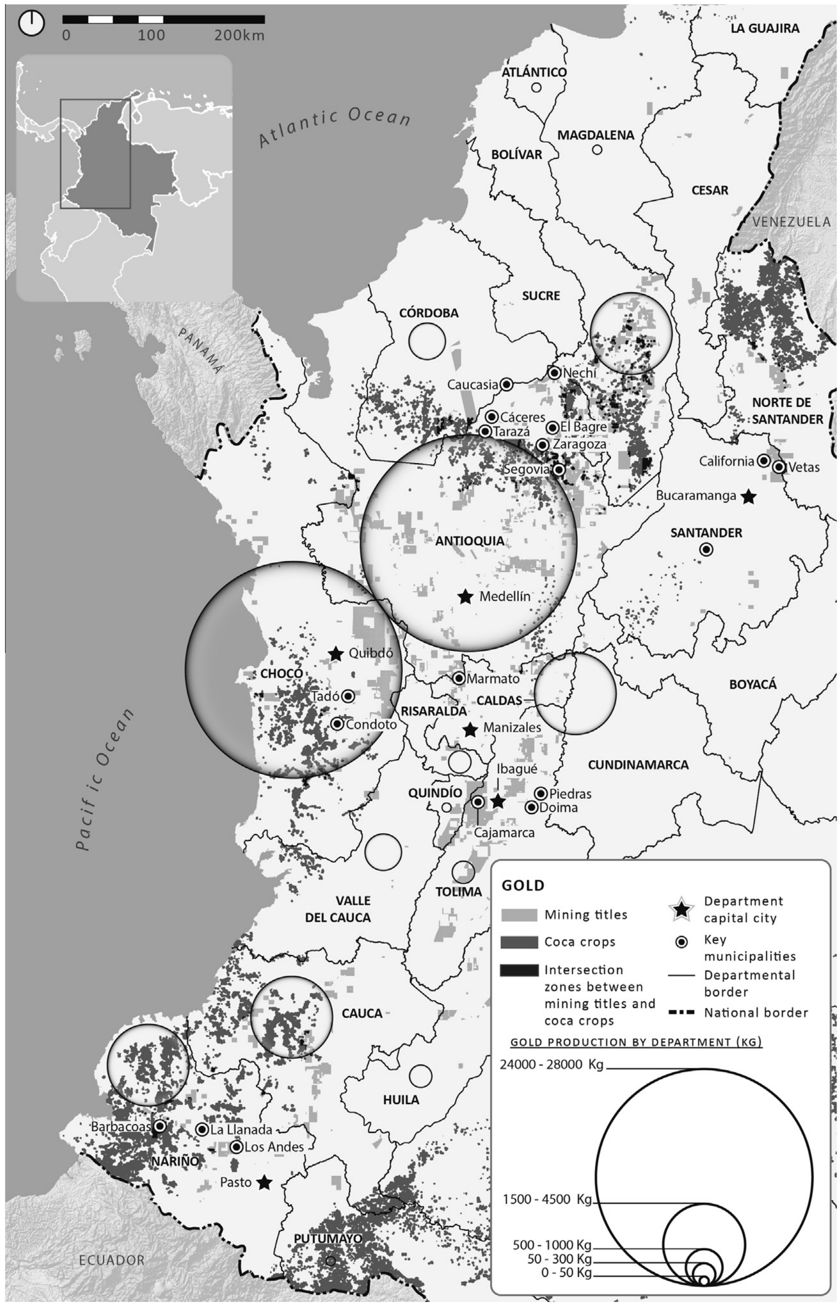
One of the more frustrating lessons for scholars and policymakers in the realm of conflict and peace studies has been the pervasiveness and resilience of the links between resources and conflict, as well as the apparent difficulty of governing and controlling illicit markets that pose risks to peace, security, and development consolidation in transitional contexts, even in the aftermath of actual fighting (Gates, Hegre, Nygård, & Strand, 2012). Indeed, much of the focus on post-conflict urban and rural security (Kunkeler & Peters, 2011; Moser & McIlwaine, 2001; Muggah, 2009; Yassin, 2008) can be traced back to findings illustrating that, at war's end (Paris, 2004), the enduring circulation of weapons combined with accumulated criminal expertise and the ongoing operation of illicit markets produce new and sometimes escalating forms of violence and crime.

Scholars have been studying the mechanisms linking resources to armed conflict for more than 15 years. Central questions addressed by the literature have revolved around the type of resources fueling civil wars and on the type of institutions that govern production and extraction of different resources (Bannon & Collier, 2003; Berdal & Malone, 2000; Collier & Hoeffler, 2004; Elbadawi & Sambanis, 2002; Fearon, 2005; Keen, 1998; Ross, 2003). In a seminal work, Ross (2004) suggested that "oil, nonfuel minerals, and drugs are causally linked to conflict, but legal agricultural commodities are not" (Ross, 2004, p. 35). In addition, the literature has adopted a term coined by Ross (2004), which refers to the lootability of resources: Lootable resources are those that can be easily transported and distributed, such as diamonds and alluvial (or placer) gold extracted in small-scale operations, which combine high value with small volume. Resources such as coal or oil, in contrast, face high barriers to entry for looters because they are only lucrative in large quantities and, in the case of large-scale mining processes, require significant investment in advanced technology.

Others have suggested that the difference among resources in terms of their contribution to armed conflict may lie in whether they are labor or capital intensive (DiJohn, 2006; Dube & Vargas, 2006). Le Billon (2001) focused on geography and proposed spatial reasons—proximity to urban centers or the capital (the closer the better protected a resource is from looting)—as well as factors associated with urban or rural, dispersed or concentrated production as explanatory factors of the resource–conflict link. Humphreys (2004) suggested that countries dependent on agricultural goods (apart from their endowment in oil or diamonds) should be considered in

Table 1. *Interviews by category, number, and percentage of total*

Actor category	Number of interviews	Percentage of interviews (%)
Mining association	14	18.18
Mining company	11	14.29
Civil society organization	10	12.99
Traditional miner	8	10.39
State official	25	32.47
Policy and academic experts	9	11.69
Total	77	



Map 1. *Gold mining titles and illicit crops in Colombia, 2014. Source: Data for gold production are based on the Colombian Mining Information System (Sistema de Información Minero Colombiano). Data on mining titles were provided by the Colombian Geological Service (Servicio Geológico Colombiano) and the National Mining Agency (Agencia Nacional de Minería). Data for illicit crops are based on the United Nations Office on Drugs and Crime (UNODC) illicit crops census (UNODC, 2013b).*

greater risk of armed conflict and should therefore concentrate on strategies of economic diversification. DiJohn (2006) proposed that high barriers to entry—and the degree to which production occurs in an enclave setting—may explain differences between resources' links to war.

The institutional context has obtained increased prominence as an explanatory factor. The most prevalent argument is that state weakness or incapacity—the lack of strong institutions—increases the likelihood for resources to become linked to political instability and corruption (Collier & Hoeffler, 2005; Karl, 1997). In line with this argument, Robinson, Torvik, and Verdier (2006) suggest that institutions that promote accountability and state competence can protect countries from a resource curse. Several attempts to distinguish among types of institutions most favorable to severing the resource–conflict link have provided important insights: Mehlum, Moene, and Torvik (2006) suggest that "grabber-friendly" institutions lower countries' income whereas producer-friendly institutions increase it. According to Snyder (2006), the relationship between lootable resources and political institutions does not necessarily lead to disorder, as long as institutions governing the extraction of resources provide room for negotiation among involved actors. Finally, Humphreys (2004) suggests that the impact of resources on conflict cannot be solely attributed to state weakness. In fact, according to Humphreys (2004), the impact of natural resources may even be independent of state strength. As a result of this institutional emphasis, natural resource management and resource extraction institutions figure prominently among the tasks of post-conflict states (Lujala & Rustad, 2011; Mähler, Shabafrouz, & Strüver, 2011; Nichols, Lujala, & Bruch, 2011), especially since natural resources have been significantly linked to conflict recurrence (Rustad & Binningsbø, 2012).

The academic literature has made important progress in improving our understanding of why people and the media tend to respond to news of resource discoveries with concern and dismay instead of optimism. But the fundamental question remains open: Why is it so difficult to cut the resou rce–conflict–crime link?

Part of the answer may lie in the insufficient attention paid to institutional inertia and learning processes among actors and institutions making the resource–conflict link resistant to change. On the one hand, the literature tends to focus on one resource at a time, mainly for methodological reasons. In doing so, little consideration is given to the wider context in which illegal and criminal actors and organizations resort to many different resources, occasionally supplementing income from one resource with income from others in order to sustain and grow their organizations and personal fortunes. A better understanding of how actors design their resource portfolios, moving back and forth between different resources and making them interchangeable as sources of revenue, can shed light on the workings of the resource–conflict link. On the other hand, the literature pays little attention to sub-national differences in the resource–conflict link. Yet some evidence suggests that, even holding the type of resource constant, local and regional contexts within conflict-torn countries vary (Koubi, Spilker, Böhmelt, & Bernauer, 2014; Leiteritz, Nasi, & Rettberg, 2009; Libman, 2010; Rettberg *et al.*, 2011; Rustad, Buhaug, Falch, & Gates, 2011). These sub-national variations in the intensity and forms of violence linked to different types of resources have not been sufficiently explored in the scholarly literature and may provide important hints for policymakers as well. Finally, the literature is tilted in favor of economic considerations (such as rent-seeking or youth unemployment) as the main drivers of illegal actor strategies. A focus on resource portfolios may shed light on additional factors at work, such as political motivations in abandoning discredited sources of funding—such as kidnappings (or the trade in rare species and ivory; see, for example, Keane, Jones, Edwards-Jones, & Milner-Gulland, 2008), reinforcing the political aspect of political economy approaches.

Many of these insights can be applied to the academic literature on the Colombian conflict and crime, which has tended to focus on the drug trade as the main driving force (Angrist & Kugler, 2008; Bagley, 2001; Holmes, Gutiérrez de Piñeres, & Curtin, 2006; IEPRI, 2006; Mejía, Restrepo, & Rozo, 2014; Pardo, 2000), neglecting attention to other resources that have become deeply enmeshed in conflict and criminal dynamics. Some efforts have sought to overcome this neglect, by exploring the links between other resources, the armed conflict, and crime (Leiteritz *et al.*, 2009; Rettberg *et al.*, 2011) and by describing the different ways and strategies through which illegal actors—not only those with a particular political agenda—try to capture the rents generated by the extractive sector (Massé & Camargo, 2012). In light of increasing information on the impact of the Colombian mining boom on illicit markets, some of this work has begun to focus specifically on gold, a legal resource (inasmuch as there are official and regulated markets for its trade and rules for its exploitation). For example, Idrobo *et al.* (2013) sought to statistically identify the impact of illegal gold mining on conflict, and found that, against a background of a national drop in homicides (Restrepo & Aponte, 2009), municipalities linked to gold-mining present higher overall homicide rates. Rettberg *et al.* (2013) provide a detailed account of local authorities' opinions and perceptions of the impact of gold mining in ten Colombian gold mining regions. Lobo and Segura (2010) provide a detailed case study of gold mining in the Chocó region, illustrating how Afro-Colombian communities clash with other groups over extraction rights and practices. Giraldo and Muñoz (2012) and Giraldo (2013) develop an in-depth study of Antioquia region to show the manifold and complicated relations between gold mining and illegal actors.

This paper intends to complement these contributions. In line with the voids in the literature on the political economy of armed conflict and peacebuilding listed above, the paper seeks to develop a better comprehension of a resource different from drugs—gold—and its links to conflict and crime at a sub-national level in Colombia, explore the notion of a resource portfolio with a focus on the links between gold and drugs, and analyze the interaction of economic and political motivations in the production of resource portfolios across time.

3. THE RELATION BETWEEN GOLD MINING, DRUGS, ARMED CONFLICT, AND CRIME IN COLOMBIA

In this section we elucidate and analyze the connections between gold mining, the drug trade, armed conflict, and crime in Colombia by tracing them at different stages of the gold mining process: exploration, extraction, refining, and commercialization. We classify the impacts of gold mining to conflict according to two broad categories: direct or indirect involvement by illegal actors. At each stage, we describe the various connections between the mining process, the drug trade, social violence, and illegal actors, in order to illustrate not only that illegal and criminal actors maintain a resource portfolio, whereby they combine multiple strategies to loot different resources at once to satisfy their funding needs, but also that a focus on the main resource fueling the Colombian wa r—drugs—overshadows the spillover of criminal activity into other licit and illicit markets. This, as will be concluded below, poses problems for policymakers seeking to consolidate peacebuilding efforts in the country. We present evidence from multiple Colombian gold-mining regions in order to illustrate the specific ways in which the relationship between gold mining and drugs has varied in different parts of the country. We begin, however, by presenting some background information on the development of the Colombian (gold) mining sector.

(a) *Mining in Colombia*

Implemented in 2001, a new mining policy, aimed at fostering private investment in mining projects, achieved impressive results (Cárdenas, Ortiz-Riomalo, & Rettberg, 2013): As illustrated by Figures 1–3, foreign direct investment (FDI) in the mining sector as well as the number of mining titles have sharply increased in the past 10 years, aided by a surge in commodity prices and by a significant improvement of public security in the country. In a period of 10 years, the country's homicide rate per 100,000 inhabitants plunged from 69.8 in 2002 to 32.3 in 2012; kidnappings, too, diminished noticeably

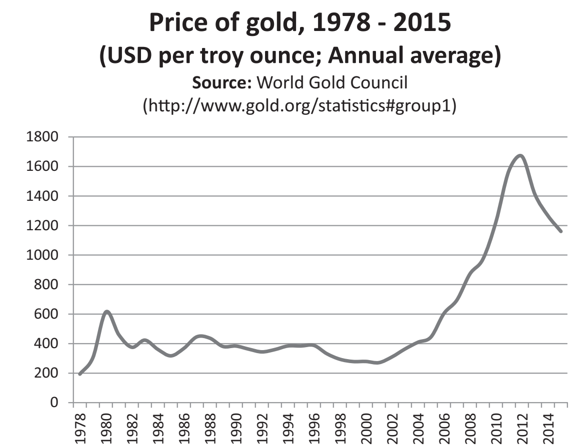


Figure 1. *Gold price 1978–2015.*

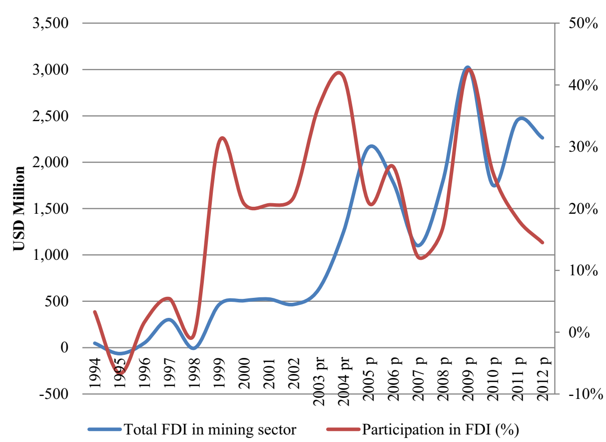


Figure 2. *Foreign Direct Investment (FDI) in mining in Colombia. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.) Source: Banco de la República. Elaborated by authors.*

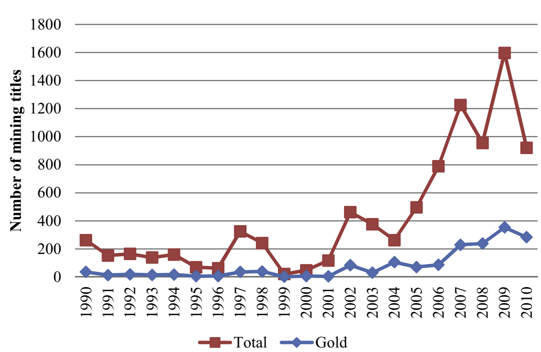


Figure 3. *Mining titles granted by Colombian Government. *Before the moratorium in new licences (April 2011–May 2013). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.) Source: Ingeominas (2012). Elaborated by author.*

(Massé & Camargo, 2012; Mejía, 2012; Ministerio de Defensa Nacional, 2010, 2013), giving rise to what the *Financial Times* has coined "the new Colombia" (*Financial Times*, 2013).

At the same time, the institutional framework adopted in 2001, while successful in attracting investors, was flawed in aspects related to the protection of the environment and the technical requirements—from accounting to equipment—for the initiation of sound mining projects. In addition, several loopholes in the distribution of mining-related royalties to municipalities created administrative gray areas, resulting in the evasion of regulatory responsibilities and opportunities for corruption (Giraldo, 2013; Giraldo & Muñoz, 2012; Rettberg & Ortiz-Riomalo, 2013). More importantly, transitional measures approved in order to legalize traditional artisanal, small- and medium-scale operations systematically failed. In 2010, according to the last Mining Census conducted so far, 63% of all Productive Mining Units in Colombia were running their operations without complying with minimum legal requirements, a figure that increases to 86% when considering only gold mining operations. In 2011 no more than the 12% of the total gold produced in Colombia was extracted by large-scale, legal and formal mining operations.[3] One mining expert involved in this process in the early 2000s described this situation as follows: "We did not create the institutions for regulation, we didn't think about those" (academic expert, authors' translation, 2013).

Institutional weakness proved to be no problem—at least from a security point of view—as long as the price of gold was low. However, the significant price hike experienced by gold during 2001–12 (Figure 1) and the increasing number of potential miners—not only mining companies—it attracted to mining municipalities has led to severe turmoil.[4] Traditional gold-mining communities vie over existing resources with newcomers, communities dedicated to agriculture have opposed the use of water by newly sprung up mining operations, and the ownership of land and mining titles has come under increasing dispute. Phenomena typically linked to enclave economies—price inflation, prostitution, corruption, and crime—have also escalated in gold-mining municipalities (Rettberg *et al.*, 2013). In the words of the mayor of one small municipality, "anyone who feels like mining can do so without any control" (Chocó), illustrating the consequences of both deficient and insufficient institutional leadership and capacity.

In addition to this social unrest, illegal armed actors—mainly the Leftist *Fuerzas Armadas Revolucionarias de Colombia* (FARC), the *Ejército de Liberación Nacional* (ELN), and the so-called criminal bands (a *melange* of renegade demobilized paramilitary and guerrilla fighters and "plain" new criminals who operate in over 20 of the country's departments)—have taken advantage of this opportunity provided by institutional weakness and increasing prices. Both as a new source of revenue and to supplement dwindling income from the drug trade and from kidnappings (see next section), these actors have developed strategies to loot the resource as well as the rents that it produces. Some of these actors already had links to the drug trade and complement their activities in illicit crops with investments in gold mining. One official in Chocó department summed up the situation as follows: "Mining brings all of those things, it has brought us criminal groups, criminal bands, guerrillas, and common theft. All the robberies, fights, criminal groups, all that has been brought by mining, they all steal from the same mine. It has even brought trouble for families and neighbors" (public official in Chocó department, authors' translation, cited in Rettberg *et al.*, 2013). "Everybody was taken off guard, including the national authorities", said the leader of a national mining organization (leader mining organization, authors' translation, 2013).

The violence associated with these dynamics has frequently made newspaper headlines and has brought back memories of the worst days of the illegal drug trade, when small groups—or criminal gangs—brutally competed over drug riches needed by armed organizations seeking to control territory. As a result, public officials and scholars have suggested that illegal gold mining is a substitute for the drug trade as a driver of violence in Colombia. The national government has even publicly declared a war on illegal mining (Presidencia de la República, 2014). Not surprisingly, while the country has experienced an overall decline in homicides, homicide rates in gold-mining municipalities have increased (Idrobo *et al*., 2013). While in non-gold-mining municipalities the homicide rate fell from about 40 homicides per 100,000 inhabitants in 2007 to less than 30 in 2010, in gold-mining municipalities this rate increased to about 45 homicides in 2008 and about 37 homicides in 2012.

Authorities have responded to the rise of illegal mining either with administrative lethargy (the application process for the formalization of mining titles has stalled since 2012, resulting in 80% of gold mining firms operating without the appropriate titles, according to the 2010 National Mining Census; see also Rojas, 2012) or with repression (a 2012 presidential decree requires the National Police to destroy machinery in use by illegal mining operations) (Ministerio de Defensa, 2012). One official at a municipal office put it bluntly: "This is war, this is innovation. We are not dealing with apprentices" (national state official, authors' translation, 2013). Both attitudes have caused further evasion of the law and resentment.

The large-scale industry has also resented the way in which the authorities' failure to respond has harmed the industry's prospects and reputation. "There is no long term mining policy, we´re being very reactive" the leader of a mining organization said, adding that "the sector's reputation is marked by illegality and informality; this is not fair" (leader mining organization, authors' translation, 2013).

Institutional weakness related to resource extraction in mining regions can therefore clearly be linked to the draining of resources into the coffers of illegal and criminal organizations.

(b) *Recent transformations of the Colombian drug trade*

Colombia has been plagued by drug-related violence since the 1970s. The pervasive effects of drugs on politics, society, and the economy have been well documented, and include corruption, institutional atrophy, and a generalized perception of state incapacity. In addition to the drug cartels, Colombian insurgents and paramilitary groups became deeply involved in the drug trade since the 1980s and 1990s (before, guerrillas had persecuted peasants involved in illicit crops for betraying the cause of the peasant revolution; see Puentes, 2006). Most recently, evidence suggests that armed actors, druglords, and political and economic elites have collaborated at different stages of the drug business in several Colombian regions.

Since consumption of illegal drugs takes place mainly in the US and in Europe (although domestic consumption has begun to increase in Colombia), the "war on drugs"—a systemic response led by the US with the collaboration of the United Nations and other governments against illicit crops in several Andean countries, which includes crop substitution and destruction efforts, interdiction of shipments, and extradition of traffickers to the US for prosecution [5]—has significantly shaped Colombia's foreign relations over several decades. Government after government has invested financial, political, and physical resources in attempts to stem the flow of drugs to the US and Europe, to curtail their political influence, and to counteract the expansion and increasing efficiency of illicit crops all over the national territory (Gaviria & Mejía, 2011). Frustration over the lack of results, however, has mounted over the years, as violence and crime linked to the drug economy have not receded, and consumption has also not declined (Reuter & Trautmann, 2009; GCDP, 2011; UNODC, 2013a). [6] At present, not only critics on the fringes of the system but former and current heads of state and mainstream public opinion share the perception that the war on drugs has reached its limits (Gaviria & Mejía, 2011; GCDP, 2011; Tickner & Cepeda, 2011) and that a fundamental shift must occur to address drugs as a public health *and* security problem (GCDP, 2011).

Regarding Colombia, a recent report by the United Nations Office on Drugs and Crime at Colombia (UNODC, 2013b) reported that the number of hectares cultivated with illicit crops has finally begun to shrink. [7] Touted by some as the victory of aerial aspersion and manual eradication [8] the report included a cautious note, suggesting that coca growers as well as illicit crop-dependent illegal groups are migrating to illegal gold mining (UNODC, 2013b).

Gold brings together many advantages in relation to illicit crops: First of all, it is a legal resource, meaning that its extraction and commercialization are regulated by domestic and international institutions, and people are not penalized for its production. In addition, gold is highly tradable as it serves as an internationally renowned asset with an externally fixed price and globally competitive markets and which plays the role of safe haven and store of value (Campodónico & Ortiz, 2002; Melo, 1974; O'Callaghan, 1993; Roache & Rossi, 2009; Sánchez, Ortiz, & Moussa, 1998). Like diamonds, gold is highly lootable, combining high value in small volume, which facilitates transport and smuggling for illegal groups. Fourth, the production of gold provides municipalities with access to public royalties, which acts as an incentive for illegal actors to loot state resources. Finally, its price experienced a steep increase for over a decade (see Figure 1). For all these reasons, gold has become an attractive alternative for illegal actors seeking to expand their revenue-generating activities in the context of increased pressure against illicit crops, their previous source of income.

Stressing the shortcomings of the war on drugs overshadows the sophisticated strategies and structures of collaboration developed in the meantime by illegal actors to mitigate the impact of aerial aspersion or manual eradication of illicit crops by diversifying their revenue-generating activities. The shortcomings of the war on drugs should thus be read not only in terms of its failure to stem the flow of drugs on a global scale, but also to prevent or reduce drug-related violence, crime and spill-over into other illicit markets within drug-producing countries.

Other measures adopted by the Colombian state to reduce violence indicators have also promoted income diversification by illegal groups. Kidnappings, for example, were for long the main supplementary source of income for illegal organizations, second only to the drug trade. This granted Colombia a well-deserved standing as one of the most insecure countries in the world. Both the reputational cost for illegal organizations as well as increased military pressure have caused this phenomenon to decline over the past years, further forcing organizations to explore alternative sources of revenue (Ministerio de Defensa Nacional, 2013).

In the next part we detail the multiple connections between gold mining, the drug trade, conflict, and crime.

(c) *Tracing the connections between gold mining, drugs, armed conflict, and crime*

Lavandaio (2008) divides large-scale mining activity into two general phases, one termed "pre-investment", and a second one in which actual extraction takes place. The first stage requires significant technological capabilities to identify the location, volume, and quality of minerals. Although no actual mining takes place, the actors involved require investment capacity and willingness to accept sunk costs, as more than ten years may elapse before projects yield returns. This explains why large-scale mining involves high barriers to entry. Only 10% of exploratory activities reach the following stage of actual exploitation (Lavandaio, 2008), involving the development of an extraction, processing, and commercial infrastructure. This second phase, called "investment", comprises mining, refining, and commercialization.

As was noted above, most ongoing mining activity in Colombia is not of this large-scale kind (DP, 2010), although most recent FDI in mining aims to develop large-scale activity. Current mining operations in gold are mainly of two kinds: *alluvial* or *placer*—manually or mechanically "washing" or dragging gold pebbles and dust out of riverbeds with reported mineral content—or *vein* or *lode* kind—by digging holes and tunnels into mountains following the actual trace of gold veins in the rock. These are small-to-medium-size operations, which are more labor intensive, require considerably lower investment capacity than large-scale endeavors, and, therefore, pose low barriers to entry. Small-scale operations also represent higher risks in terms of health and safety for workers [9] and the environment and are less efficient in the recovery of mineral from the ground (between 20% and 60%, in contrast with around 90% for large-scale operations).

The types of economic actors participating in the mining business at each of these stages and operations vary significantly: Whereas large-scale operations are usually conducted by large, formal, often multinational companies with significant investment and technological capacity as well as the ability to transfer expertise and knowledge from one geological location to another, small-scale operations are dominated by local, often informal actors who are highly vulnerable to price fluctuations and may join or abandon the activity with relative ease, employ less sophisticated instruments and infrastructure to obtain the metal, and require less capital to operate.

For the purposes of our analysis the gold extraction process will be divided into three phases: (1) exploration, (2) extraction and transformation, and (3) commercialization. Each phase offers particular opportunities for illegal and criminal actors interested in looting the resource or the wealth it generates. However, the most critical phase in terms of vulnerability to criminal activity is the extraction and transformation phase, when gold is recovered from the ground, transformed, and sold. It is also in this stage where the connections between illegal and informal gold mining and the drug trade become most apparent, as will be shown below.

(i) *Exploration*

Exploration is mainly reserved to large-scale operations. Small-scale operations also have an exploration phase, but focus on local knowledge and reconnaissance rather than geological research. Exploration companies and activities have been subjected to different levels of extortion by illegal actors who demand to be paid in order to allow the onset or continuation of exploratory and extractive, and punish the failure to comply through various acts of sabotage (from destruction of infrastructure to kidnapping of company personnel or subcontractors). This is comparable to what happens in other extractive industries, such as oil, in which different illegal actors—including guerrillas and criminal bands—have been shown to direct attacks against operations for financial and ideological reasons. Mining companies have thus adopted similar protection strategies, for instance, the signing of security agreements with military and Police forces and the hiring of private security companies (Massé & Camargo, 2012).

In this phase, also, mining companies have faced important opposition in several regions. Where agriculture has historically been the main economic activity or where strategic ecosystems like Andean wetlands have been located, mining has been questioned for its environmental, social, and labor impacts. Officials in Santander expressed concerns about toxic waste in the water flowing out of the mines (public official, 2013), while agricultural communities in Tolima rallied against gold mining for fear of lacking insufficient access (leader, mining organization, 2013). In fact, as suggested by the Center for Popular Education and Research—Peace Program social conflicts or mobilizations related to mining have sharply increased since 2008 (CINEP/PP, 2012). [10]

In sum, during the exploration or pre-investment stage gold mining may contribute to conflict by funding illegal actors—many of them with links to drugs—as well as by stirring up social conflicts, which, in some cases, have been permeated or instrumentalized by armed actors.

(ii) *Mining and refining*

The bulk of the direct and indirect links between illegal gold mining, the drug trade, and crime, both in terms of diversity of connections and value of the business, takes place in the extraction and transformation phase, which involves both large- and small-scale operations (Rettberg & Ortiz-Riomalo, 2013). This is the most labor-intensive phase of the gold mining process, as well as the one in which more people are vulnerable to becoming connected to criminal activities (Giraldo & Muñoz, 2012; Massé & Camargo, 2012), in part due to the many institutional and regulatory voids in mining activities, which compound the opportunities for illegal actors to penetrate the business. As described by the leader of a mining organization, "the country was unprepared for the mining boom, both institutionally as well as from a technical point of view" (leader of mining organization, authors' translation, 2013).

Indirect links between illegal gold mining and criminal activity are the most prevalent and include some of the following: Much like during the exploration phase, operations are subject to protection payments. "It's easier to extort gold than to grow coca", one Police official said (Police official, authors' translation, 2013). Although paying extortions and protection money to illegal organizations has been harshly denounced by the government as a measure that, in the end, serves to strengthen illegal organizations' financing structures and overall viability (Presidencia de la República, 2014), in practice the state has limited capacity to enforce non-payment. State inaction thus makes companies and individual miners more vulnerable and hence more likely to give into pressure from illegal groups or seek non-legal forms of protection, a finding that was confirmed to us by all interviewees, including national and local state officials, mining organizations and miners, and academic and policy experts.

In the case of small- and medium-scale operations, entry into mines as well as earnings have been illegally taxed by criminal bands, the paramilitary, and guerrilla groups, who in some cases charge up to 50% of each mine's earnings. The prototypical case illustrating this close relationship between a resource, conflict, and crime is the Bajo Cauca sub-region of the Antioquia department (center of the country, see Map 1), which has been controlled for decades by druglords and paramilitary groups linked to the illegal drug trade. Gold mining has been historically practiced in municipalities such as Segovia, Buriticá, and El Bagre. One large gold-mining company, the Frontino Gold Mining Company, now owned by Gran Colombia Gold, for long dominated extraction, along with the only Colombian-owned gold extraction company, Mineros S.A. Based on interviews with mine workers and local officials, and in the absence of formal data provided by the groups themselves, we estimate that any one armed group could be earning between USD 8,000 and 80,000 per operation per year from looting gold mining operations—most of them informal, illegal, and small- or medium-scale operations. An armed group could be collecting between one million and 10 million USD per year. [11] The amount of money that illegal armed actors extract from mining operations can increase to the extent that such operations are usually looted by more than one actor (between one and four actors on average), as becomes clear in the following testimony of a miner in Antioquia:

> "Monthly we [the miners] have to pay extortions to [*all the armed groups operating in the zone*]. We also have to pay the Police...they bribe us to let us transport fuel for machinery or explosives and to let us continue running our operations."
>
> [Mine worker of Bajo Cauca, Antioquia, September 2012]

In this sense, mining operations are vulnerable not only to illegal armed groups but to corrupt state officials, as well. In addition, miners and illegal actors have collaborated with armed actors or bribed the military or the Police in order to obtain access to hard-to-come-by supplies such as explosives (used to destroy the mineral-containing rock and officially sold only by the Colombian state, which restricts and controls uses and circulation), mercury and cyanide (used to separate the metal from other elements in the rock), and diverse types of equipment, such as dredges and mechanical diggers of different sizes and costs. In order to dissuade state authorities from enforcing mining or environmental laws, miners have even collaborated with armed actors to intimidate authorities, providing an opportunity for social unrest and protest to become intertwined with the dynamics of crime and armed conflict. Interviews conducted in the Nariño department (South of the country, see Map 1), for example, suggest that local guerrilla commanders were asked by locals to exert pressure on the regional environmental management center so as to prevent the closing of several informal mining operations. It appears that state authorities—mainly Police forces—collude with illegal actors in several regions of the country to demand a cut of gold production profits or charge protection payments to producers.

Illegal actors have also participated in gold extraction in more direct ways. This has taken the form of directly running gold extraction and transformation operations or forming informal alliances with mine workers to conduct extraction and transformation. In these cases, armed actors play a role in controlling mines, managing workers, and investing in capital and equipment. This situation—which has been reported in the south of the department of Tolima, the north and northeast of the department of Antioquia, and the Pacific region (Contraloría General de la República, 2012; Maldonado & Rozo, 2014; Policía Nacional de Colombia, 2013; Romero *et al.*, 2014), among others—is less frequent than indirect looting primarily because illegal actors tend to lack the experience and expertise required to run extractive operations directly. As a result, they opt for the most cost-effective way to obtain income from a resource, which is to syphon off the work of others. Some actors have created fake companies in order to legalize their participation in mining operations and skim off part of their profits.

The price hike experienced by gold during 2001–12 (Figure 1), combined with increased pressure on illicit crops via manual eradication and aerial aspersion efforts, has led to the existence of a redundant workforce in illicit crops which has been absorbed by gold mining activity (UNODC, 2013b). This migration from one resource to another has been reported in the south of the country (Nariño and Putumayo departments) and in the Bajo Cauca region of Antioquia (Map 1), and is likely taking place elsewhere. It is most noticeable in the short and black-stained nails of former coca harvesters (*raspachines*), who scrape off coca leaves from plants, and are now dedicated to mining. According to one public official in the Mayor's Office of Los Andes Sotomayor, in Nariño region (see Map 1), people in the region have worked interchangeably in gold mining and in illicit crops. Pointing to the coffee fields next to the refining machinery, he said "two years ago, this was all coca bushes" (public official, authors' translation, 2013). Fluctuations in workers' involvement in mining or other activities depend both on whether the price of gold makes mining worthwhile or whether government-led eradication efforts impede illicit crop cultivation, as has been the case in recent years.

In addition to peasants' and workers' economic migration described above, Police and judicial authorities have gathered evidence—based on information stored on confiscated computers and provided by demobilized fighters—indicating that commanders of both guerrilla and criminal bands have issued explicit orders and strategies to gain access to gold mines and miners (InSight Crime, July 26, 2013). Their goal is to complement dwindling resources from illicit crops and to engage in cooperation with other illegal actors operating in the same region in search of mutual gains. As becomes evident from Map 1, this purpose is aided by the physical proximity of illicit crops and mining activities in many Colombian regions.

In brief, in the mining and refining stage, different illegal actors—most of them with ties to the drug trade—loot gold mining, especially by targeting small-scale operations, which are especially vulnerable. In some cases, illegal actors regulate and manage the mining process by allocating capital, labor, and inputs in order to get a cut of the profits. In a few cases illegal actors appear to be directly running mines. The connections with the drug trade throughout this phase are numerous and diverse: Illegal gold mining has absorbed a redundant workforce following increased pressure on illicit crops throughout the country. In addition, some of the illegal groups controlling entry to mines and management of output maintain parallel operations aimed at exacting income from illicit crops, either because mines and illicit crops are physically close and under the same territorial control, because the price hike has favored the incursion into new resources, or because illegal organizations have designed specific strategies to venture into new regional economies. Gold and drugs have thereby become deeply enmeshed as resources fueling conflict and crime, suggesting the existence of resource portfolios developed by illegal actors to support their activities.

(iii) *Commercialization*

Many of these connections carry through to the last phase of the gold mining process, when gold is sold to intermediaries and to Colombia's Central Bank, which has offices in the main mining regions (mainly in Medellín and Quibdó, the capitals of the Antioquia and Chocó departments) and in Bogotá, the national capital. It is at this stage when official gold production is calculated. And it is here when gold enters international markets, from jewelry to financial values and bonds. For illegal actors, what is at stake at this stage no longer refers to the value of the resource alone, but its nature as an instrument toward other goals. Drug lords, for example, have seized the opportunity provided by booming yet under regulated mining markets to launder drug-related revenues in collaboration with willing or subservient authorities. This is suggested by evidence about drug traffickers buying gold in countries such as Panama and Mexico or in other Colombian regions and smuggling it to "friendly" municipalities (sometimes with the participation of legally established gold-trading companies). Once the gold is brought to their municipality, local officials will report it as produced in their area of influence, claim royalties on the reported gold, and then share earnings with drug traffickers (Defensoría del Pueblo, 2010, pp. 26–27). One public official from Chocó described the situation:

> Gold arrives in a commercial flight, not as a remittance, and these people in bulletproof cars arrive and pick it up. (. . .) The gold gets traded, they bring documents, the documents are nothing, I have looked into this. The gold leaves the airport with a document saying ``I´m sending 6,000 g of gold to Medellín" and later some company says it is sending a certain amount of money as a result of gold trading, but no one checks, no one knows, no one says anything. (. . .) So there can be asset laundering, because there is no effective control.
>
> [public official, authors' translation, Chocó, 2013]

This is also suggested by the fact that the volume of exported gold tends to exceed registered production (Uribe, 2014). Local authorities, in turn, seek to attract gold sellers from other municipalities in order to report their gold as having been produced locally and then obtain royalties from it (Contraloría General de la República, 2012; Uribe, 2014).

Only this can explain the case of Maceo, a town that consistently reports production and claims royalties but in which not only no mining titles have been distributed but also no known mining operations are taking place (Giraldo & Muñoz, 2012). More than 35 other municipalities are under investigation for similar practices (Contraloría General de la República, 2013). Corruption at the government level also happens at the stage of royalty distribution. As in the case of other natural resources such as oil, elected officials award contracts under the guise of providing public services, a portion of which is then used either for private purposes or paid out to illegal actors exerting armed pressure on officials.

Authorities seeking to control these dynamics, however, do not seem to have been able to keep track of developments. As described by one public official in Antioquia:

> We have an office dedicated to mining, and, I ask, what does gold have to do with asset laundering? Only now are we beginning to ask the right questions, so when someone says ``there´s asset laundering in gold mining", who is following that money? Who owns those machines? We are still in diapers, we have a growing problem and no regulatory backing. All the lessons we learned from the drug trade have not been translated here, so we are reinventing everything.
>
> [Interview, public official, Antioquia 2013]

Table 2 synthetizes the links through which gold mining is directly or indirectly related to armed conflict, crime, and the drug trade in Colombia.

4. DISCUSSION

"You wish you had a magical wand and, 'plush', you could just make them disappear" said an Antioquia-based state official, referring to the consequences of gold mining, armed conflict, and the drug trade in his region (cited in Rettberg *et al*., 2013). Our analysis illustrates the difficulties of overcoming these predicaments, as a result of the pervasiveness and resilience of the links between resources and conflict. Illicit crops and gold mines are not only physically close, but also share many links with illegal armed actors. As a result, our study puts in perspective the overall gains in security achieved by Colombia in national terms, and reinforces the need for a

Table 2. *Direct and indirect connections between illegal gold mining, the drug trade, and criminal actors, according to the phases of the gold extraction process*

	Pre-investment	Investment	
	Exploration	Mining and Refining	Commercialization
Indirect	(1) Extortion by illegal actors of companies involved in exploration, mainly linked to large companies	(1) Illegally taxing mines	
	(2) Sabotage of mining operations to induce extortion and protection payments	(2) Collaboration with armed actors and official forces for access to supplies (explosives, mercury, cyanide, and equipment)	
	(3) Social unrest caused by discussion of feasibility of mining operations	(3) Collaboration with armed actors to dissuade or intimidate authorities from enforcing the law	
		(4) Absorption of residual workforce and capital from illicit crops	
Direct		(1) Operation and management of illegal mines	(1) Drug-related money laundering via collaboration with local authorities
		(2) Participation in profits from legal operations	(2) Corrupt use of gold-related royalties.
		(3) Acquisition of mines for laundering drug-related money	

Note: Shaded areas refer to specific connections with the drug trade.
Source: Elaborated by authors based on secondary sources, interviews, and fieldwork.

sub-national focus in order to fully capture the realities of armed conflict, the political and economic dynamics sustaining conflict and crime in different regions, and the perspectives of building durable peace.

Our study both vindicates and complements some of the central tenets of the literature on the political economy of armed conflict and peacebuilding. Generally speaking, in line with this literature, gold mining operations tend to take place far from the main urban centers, in peripheral and remote areas where institutional weakness and lucrative and low-barrier resources converge to provide illegal armed actors with the motive and the opportunity to engage in systematic looting activity. In the absence of formal institutions capable of governing licit and illicit markets (Cárdenas *et al.*, 2013; Garay, 2013), disputes over territorial control and control of mining rents are solved among the vying groups, which accounts for much of the violence linked to gold mining (Idrobo *et al.*, 2013).

In addition, the kind of small-scale and informal mining prevailing in Colombia poses low barriers to entry: Required qualifications are few, as are techniques and investments (both human and physical). Time horizons are shorter overall than in large-scale mining, further lowering barriers to entry. This applies especially to the case of placer mining (in contrast with lode mining, where larger amounts of rock need to be removed, crushed, and processed before accessing the actual metal). The lower efforts required to take part in mining are most clearly visible in the absorption by informal mining operations of redundant workers from illicit crops, who need few additional skills to scraping coca leaves.

The price hike experienced by gold over the past decade (Figure 1) provided further fertile ground for social turmoil as it raised expectations in mining communities and promoted competition within communities and among national and local authorities over the control of mineral wealth. This provided an additional opportunity for illegal actors to tap into social unrest and skim off some of the mineral wealth via the direct and indirect mechanisms described above.

Institutional weakness, low qualifications, and low barriers to entry make small-scale mining highly vulnerable to looting (as well as perpetuated by it) and attractive for illegal organizations seeking new sources of revenue. Regulation of extraction is deficient, causing civilian authorities and Police and military forces to either clash over each other's perceived responsibilities or drag their feet when it comes to rule enforcement (or plainly engage from corrupt practices). Such foot-dragging and corruption, in turn, provides opportunity for illegal and criminal actors seeking to evade both national and local state control.

When combined with a price hike (a financial incentive) as well as with growing political pressure on illegal armed groups in terms of cleaning up their financial records vis-à-vis an increasingly critical international community and the perspective of negotiating with the state, these factors can explain the growing attention that gold mining has received by these organizations all over the national territory.

Many of these findings apply to the relation between gold and illicit crops. Like mining regions, illicit crops tend to be located in marginal areas where institutions face difficulties for governing economic activity, in general, and resource extraction, in particular. Often, illicit crops are just one among other (legal and illegal) economic activities taking place, due to similar geographic and climatic requirements. When market conditions improve the profitability of a particular non-drug-related economic activity (such as a price hike in the case of gold), or when the political cost of other sources of income increases (as has been the case with kidnappings), it may be of interest for illegal armed groups to increase their participation in that other activity's value chain, participating in production or laundering their drug-related assets. In brief, the criminal infrastructure and knowhow developed by illegal armed groups to exert control and power in illicit crop production can easily be transferred to other activities when need or opportunity arise. This conforms to our proposition that illegal organizations develop a resource portfolio which guides their revenue-raising and territorial expansion strategies and which follows both financial and political considerations.

The findings confirm that institutions ruling resource extraction—especially in the mining and the commercialization phases described above—play a crucial role. In general, the institutional context where mining activity takes place conditions its relations to conflict and the way it would be permeated by conflict and illegal actors. The weaker the institutions governing a specific economic activity taking place where illegal armed groups are involved in illicit crop production and trade, the higher the likelihood for this activity to be permeated by drug traffickers and producers. This has adverse implications for efforts to consolidate state institutions as an antidote for violence and crime. As suggested by the academic literature described in Section 2, clear and enforceable rules regarding rights and methods of extraction as well as regarding the distribution of resulting wealth promote accountability, state competence, and reduced space for interference by criminal actors. Their ongoing weakness in Colombia will not be solved by an imminent peace agreement and may, instead, provide opportunities for the transformation of criminal organizations in control of resource portfolios. This is a crucial message for the policy formulation process.

5. CONCLUDING REMARKS

After all, General Naranjo may have been right when he said illegal gold mining is worse than the drug trade. Not only in causing present violence and criminality, but also in laying the seeds of future turmoil, gold mining's links to armed conflict, crime, and the drug trade as discussed here suggest that its effects may be as pervasive as those of an illegal resource. Therefore, drawing stark distinctions among resources in our understanding of the political economy of armed conflict and crime is both analytically flawed as well as futile from a policy prescription perspective.

Faced with the likelihood of an imminent negotiated end to armed conflict in Colombia, our findings provide insights into the prospects of durable peace in the country. While the termination of FARC's involvement in illicit drugs is part of the negotiation with the national government, it is clear that the drug trade cannot be stopped by decree. In addition, as has been shown here, ongoing institutional weakness and the ongoing transformation of criminal groups show that there exist multiple alternative opportunities for tapping into additional licit and illicit economies. In this sense, our focus on gold and drugs as interchangeable conflict resources is just one of many possible combinations in the design of resource portfolios supporting armed conflict or crime.

In the face of ongoing weakness in many regions of the country and sectors of society, the Colombian state and institutions provide fertile ground for the various connections of gold mining in Colombia with criminal and illegal armed actor activities, most importantly the drug trade. As commodities keep dominating the global economic agenda and as the Colombian state continues its commitment to promote mining as a strategic activity, opportunities for criminal actors seeking lucrative sources of income in designing their resource portfolios abound. If left unattended, these may significantly hamper progress toward the eventual consolidation of peace in the country.

To be sure, there are some regional variations, as some municipalities and departments have stayed clear from this gold-to-violence link (Rettberg & Ortiz-Riomalo, 2013). Ongoing research, for example, suggests that in regions in which anti-drug efforts have been most successful—i.e., the area devoted to growing coca leaves has decreased—gold mining appears to have emerged as a substitute for illicit crops, while it serves only as a complement for revenue in other regions (Mejía, Rettberg, Idrobo, & Ortiz-Riomalo, 2013). In addition, there may be differences in the gold-to-drugs-to-crime link depending on the kind of gold deposits and mining operation arrangements prevailing in specific regions, a subject that we explore in Rettberg and Ortiz-Riomalo (2013). Therefore, many of the connections between different war economies mentioned here need to be further explored, understood, and taken into account for policymaking. For example, "stick" policies are required for cutting direct mechanisms in order to delink gold mining from the drug trade, crime, and conflict. In addition, indirect mechanisms require "carrot" policies that support mining operations, manage social conflicts, and involve the relevant organizations and institutions. Of course, in the end what is needed most for stemming all kinds of violence and criminal activity in the country is territorial control and institutional strength by the Colombian state. This remains the country's most important and basic pending task.

NOTES

1. The Colombian armed conflict began in 1964, when the two main Leftist guerrillas—the *Fuerzas Armadas Revolucionarias de Colombia* (FARC) and the *Ejército de Liberación Nacional* (ELN) were formed. Both are still active today, although, as of this writing, both are involved in talks over an eventual demobilization following a peace agreement. Between the 1980s and 2006, guerrillas were confronted not only by the Colombian Armed Forces but by Rightist paramilitary forces, which have since demobilized. The conflict has claimed over 200,000 lives and has produced the forceful internal displacement of close to five million people (Cohen & Deng, 2009; Fagen, 2003; GMH, 2013; Ibáñez, 2008). Remaining FARC and ELN fighters amount to about 11,000 individuals.

2. We thank an anonymous reviewer for bringing this point to our attention.

3. In Colombia there are only two companies extracting gold—*Mineros S.A.* and *Gran Colombia Gold*. The rest of the fully legal projects in gold mining are in the exploration or pre-investment stage. Taken together, these two companies produced only 6.5 tons in 2011, in contrast with 55.1 tons of gold produced in Colombia in that same year. When considering other legal mining operations, not more than 15% of total gold produced comes from legal gold mining operations (*Dinero*, June 9, 2011).

4. The negative effect of income shocks is not restricted to minerals, as suggested by Fjelde (2015) for the case of the agricultural sector in Africa, in which lower agricultural income can be linked to increases in rebel recruitment.

5. For further information on the "war on drugs" in Colombia and the Andean region, see GCDP (2011), Lupsha (1997), Mejía and Restrepo (2011), Mejía and Gaviria (2011), Tabares and Rosales (2005), Thoumi (1997, 2002), UPDADH (1989), WOLA (1993).

6. According to the Global Commission on Drug Policy (GCDP, 2011), consumption of opiates, cocaine, and cannabis, respectively, increased 34.5%, 27% and 8.5% during 1998–2008. Reuter and Trautmann (2009) reach similar conclusions for the case of Europe during 1998–2007. In line with this point, the last World Drug Report from the United Nations Office on Drugs and Crime (UNODC, 2013a) remarks that the "drug use situation has remained stable", although "there has been some increase in the estimated total number of users of any illicit substance". Violence associated with drug trafficking has not disappeared either. On the contrary, in regions such as Central America and Mexico it has actually increased (Reuter & Trautmann, 2009; UNODC, 2013a).

7. However, world cocaine production has remained stable (The Economist, April 2, 2013; UNODC, 2013a). Recently, Peru and Bolivia have increased their participation in global cocaine markets.

8. Aerial aspersion has been a common practice to combat illicit crops since the 1970s, giving rise to debates about its efficiency in reducing cultivated area and the risks it poses to human and animal health. Manual eradication—the physical destruction and uprooting of plants by hired workers—was added as a complementary strategy in the 1980s and 90s in order to increase the effectiveness of eradication efforts (Tabares & Rosales, 2005).

9. Health threats refer especially to mercury poisoning leading to respiratory and liver diseases and malformations (Sieber, 2013).

10. Conflicts associated with gold mining are mainly related to four issues: (1) the environment (opponents fear harm to ecosystems located near mining operations); (2) economic rivalry (fears that mining will affect the viability of agriculture or ecotourism); (3) scale of mining operations (discussions revolve around the appropriate titles and permits to allow this kind of activity); and (4) labor issues (related to salaries and benefits) (CINEP/PP, 2012).

11. This figure results from the product between the amount of money that an armed group can obtain and 125, the minimum number of gold mining operations operating in this region. We assume all operations are vulnerable to looting by illegal armed actors.

REFERENCES

Angrist, J., & Kugler, A. (2008). Rural windfall or a new resource curse? Coca, income, and civil conflict in Colombia. *The Review of Economics and Statistics, 90*(2), 191–215.

Arnson, C., & Zartman, W. (Eds.) (2005). *Rethinking the economics of war: The intersection of need, creed, and greed*. Washington, DC: Hopkins University Press.

Bagley, B. (2001). *Drug trafficking, political violence and U.S. policy in Colombia in the 1990s*. Retrieved from <http://www.ugalapagosfall.miami.edu/media/college-of-arts-and-sciences/content-assets/international-studies/documents/publications/Bagley%20Drugs%20and%20violence%20final3.pdf>.

Bannon, I., & Collier, P. (2003). Natural resources and conflict: What can we do. In *Natural resources and violent conflict: Options and actions*. Washington, DC: World Bank.

Berdal, M., & Malone, D. (2000). *Greed and grievance: Economic agendas in civil wars*. Boulder: Lynne Rienner Publishers.

Biggs, T. (2012). *Mozambique's coming natural resource boom: Expectations, vulnerabilities and policies for successful management*. Confederação das Associações Económicas de Moçambique (CTA), USAID.

Boyle, M. (2014). *Violence after war: Explaining instability in post-conflict states*. Washington, DC: JHU Press.

Campodónico, H., & Ortiz, G. (2002). *Características de la inversión y del mercado mundial de la minería a principios de la década de 2000*. Santiago de Chile: CEPAL, ECLAC, Naciones Unidas.

Caracol Radio (2012). *General Naranjo advierte que minería ilegal puede llegar a ser peor que el narcotráfico*. Retrieved August 20, 2012, from Caracol Radio web site <http://www.caracol.com.co/noticias/actualidad/general-naranjo-advierte-que-mineria-ilegal-puede-llegar-a-ser-peor-que-el-narcotrafico/20120323/nota/1659437.aspx>.

Cárdenas, J., Ortiz-Riomalo, J., & Rettberg, A. (2013). *Ni romance ni pesadilla: aproximaciones regionales a la minería de oro en Colombia* Bogotá, May 23rd–24th: Presentation delivered at the international conference "Minería en Latinoamérica: retos y oportunidades", Bogotá, September 25th–27th, Universidad de los Andes.

CINEP/Programa por la paz. (CINEP/PP) (2012). *Minería, conflictos sociales y violación de derechos humanos en Colombia*. Bogotá, D.C.: Informe especial. Octubre. CINEP/PP.

Cohen, R., & Deng, F. (2009). Mass displacement caused by conflicts and one-sided violence: National and international responses. In *SIPRI, SIPRI yearbook 2009* (pp. 15–38). Stockholm: Stockholm International Peace Research Institute.

Collier, P., & Hoeffler, A. (1998). On the economic causes of civil war. *Oxford Economic Papers, 50*, 563–573.

Collier, P., & Hoeffler, A. (2004). Greed and grievance in civil war. *Oxford Economic Papers, 56*(4), 563–595.

Collier, P., & Hoeffler, A. (2005). Resource rents, governance and conflict. *Journal of Conflict Resolution, 49*(4), 451–482.

Colombia. Ministerio de Defensa (2012). *Decreto No. 2235 en relación con el uso de maquinaria pesada y sus partes en actividades mineras sin las autorizaciones y exigencias previstas en la ley* Presidencia de la República. Retrieved from <http://wsp.presidencia.gov.co/Normativa/Decretos/2012/Documents/OCTUBRE/30/DECRETO%202235%20DEL%2030%20DE%20OCTUBRE%20DE%202012.pdf>.

Contraloría General de la República (2012). *La explotación ilícita de recursos minerales en Colombia. Casos Valle del Cauca (Río Dagua) – Chocó (Río San Juan). Efectos sociales y ambientales*. Bogotá: Contraloría General de la República, Retrieved from <http://www.contraloriagen.gov.co/documents/10136/175971301/La+Explotaci%C3%B3n+Il%C3%ADcita+de+Recursos+Minerales+en+Colombia.+Casos+Dagua+y+San+Juan.pdf/49219b35-39ed-4aa0-881f-165be0fc6871?version=1.0>.

Contraloría General de la República (2013). *Contraloría advierte posible lavado de activos a través de regalías ficticias*. , Retrieved from Contraloria General de la Nacion web site <http://www.contraloriagen.gov.co/web/guest/boletines-2013/-/asset_publisher/mQ19/content/contraloria-advierte-posible-lavado-de-activos-a-traves-de-regalias-ficticias>.

Defensoría del Pueblo (DP) (2010). *Minería de hecho en Colombia*. Bogotá, D.C.: Defensoría del Pueblo, Imprenta Nacional de Colombia.

DiJohn, J. (2006). Abundancia de Recursos Minerales y Conflicto Político Violento: Una Evaluación Crítica. In F. Gutiérrez, M. Wills, & G. Sánchez (Eds.), *Nuestra guerra sin Nombre:Colombia, Transformaciones del conflicto*. Bogotá, D.C.: Instituto de Estudios Políticos y Relaciones Internacionales.

Dinero (2011). *La nueva lavandería*. Retrieved August 13, 2012, from Dinero web site <http://www.dinero.com/caratula/edicion-impresa/articulo/la-nueva-lavanderia/120971>.

Dube, O., & Vargas, J. (2006). *Resource curse in reverse: The coffee crisis and armed conflict in Colombia*. Bogotá: Documentos CEDE, Facultad de Economía, Universidad de los Andes.

Elbadawi, I., & Sambanis, N. (2002). How much civil war will we see? Explaining the prevalence of civil war. *Journal of Conflict Resolution, 46*(3), 307–334.

Fagen, P. (2003). *International displacement in Colombia: National and international responses*. Copenhagen: Institute for International Studies.

Fearon, J. D. (2005). Primary commodities exports and civil war. *Journal of Conflict Resolution, 49*(4), 483–507.

Financial Times (2013). *The New Colombia: Peace and prosperity in sight, the country comes of age*. Retrieved from <http://www.ft.com/cms/d2ec5854-cd1e-11e2-9efe-00144feab7de.pdf>.

Fjelde, H. (2015). Farming or fighting? Agricultural price shocks and civil war in Africa. *World Development, 67*, 525–534.

Garay, L. (2013). *Minería en Colombia. Fundamentos para superar el modelo extractivista*. Bogotá, D.C.: Contraloría General de la Nación.

García, A., & Llinás, G. (2012). Los Programas de Desarrollo y Paz (PDP) de Colombia: una apuesta innovadora por el desarrollo y la paz en medio del conflicto armado. In A. Rettberg (Ed.), *Construcción de paz en Colombia* (pp. 273–304). Bogotá, D.C.: Ediciones Uniandes.

Gates, S., Hegre, H., Nygård, H., & Strand, H. (2012). Development consequences of armed conflict. *World Development, 40*(9), 1713–1722.

Gaviria, A., & Mejía, D. (Eds.) (2011). *Políticas antidroga en Colombia: éxitos, fracasos y extravíos*. Bogotá: Ediciones Uniandes.

Gelb, A., & Majerowicz, S. (2011). *Oil for Uganda – Or Ugandans? Can cash transfers prevent the resource curse?*, Working paper 261. Center for Global Development.

Giraldo, J. (2013). *El gobierno del oro en el Bajo Cauca*. Medellín: Universidad EAFIT.

Giraldo, J., Misse, M., Thoumi, F., Reno, W., Duncan, G., Tobón, S., ... Ortega, J. R. (2013). *Economía criminal y poder político*. Medellín: Universidad EAFIT.

Giraldo, J., & Muñoz, J. (2012). *Informalidad e ilegalidad en la explotación del oro y la madera en Antioquia*. Universidad EAFIT, Proantioquia.

Global Commission on Drug Policy (GCDP) (2011). War on drugs. Report of the global commission on drug policy. Retrieved from <http://issuu.com/gcdp/docs/global_commission_report_english/1?e=4620863/2855236>.

Grupo de Memoria Histórica (GMH) (2013). *¡Basta ya! Colombia: Memorias de guerra y dignidad*. Bogotá, D.C.: Centro Nacional de Memoria Histórica; Departamento para la Prosperidad Social; Presidencia de la República.

Hammond, D. S., Gond, V., de Thoisy, B., Forget, P. M., & DeDijn, B. P. (2007). Causes and consequences of a tropical forest gold rush in the Guiana Shield, South America. *American Biologist, 36*(8), 661–670.

Hilson, G., & Potter, C. (2005). Structural adjustment and subsistence industry: Artisanal gold mining in Ghana. *Development and Change, 36*(1), 103–131.

Holmes, J., Gutiérrez de Piñeres, S., & Curtin, K. (2006). Drugs, violence, and development in Colombia. A department-level analysis. *Latin American Politics and Society, 48*(3), 157–184.

Humphreys, M. (2004). Natural resources, conflict, and conflict resolution: Uncovering the mechanisms. *Journal of Conflict Resolution, 49*(4), 508–537.

Ibáñez, A. (2008). *El desplazamiento forzoso en Colombia: un camino sin retorno hacia la pobreza*. Bogotá, D.C.: Ediciones Uniandes, Centro de Estudios sobre Desarrollo Económico (CEDE) y Facultad de Economía.

Idrobo, N., Mejía, D., & Tribín, A. (2013). *Minería ilegal y violencia en Colombia* Bogotá, D.C.: Presentation delivered at the "International Conference Minería en Latinoamérica: retos y oportunidades", Bogotá May 23rd and 24th, Universidad de los Andes.

InSight Crime (2013). *ELN Profundiza su Participación en la Minería Ilegal*. Retrieved from InSight Crime web site <http://es.insightcrime.org/noticias-del-dia/eln-profundiza-su-participacion-en-la-mineria-ilegal>.

Instituto de Estudios Políticos y Relaciones Internacionales (IEPRI) (2006). *Nuestra guerra sin nombre: transformaciones del conflicto en Colombia*. Bogotá, D.C.: Universidad Nacional de Colombia.

Karl, T. L. (1997). *The paradox of plenty: Oil booms and petro-states*. University of California Press.

Keane, A., Jones, J. P. G., Edwards-Jones, G., & Milner-Gulland, E. J. (2008). The sleeping policeman: Understanding issues of enforcement and compliance in conservation. *Animal Conservation, 11*(2), 75–82.

Keen, D. (1998). *The economic functions of violence*. New York: Routledge.

Koubi, V., Spilker, G., Böhmelt, T., & Bernauer, T. (2014). Do natural resources matter for interstate and intrastate armed conflict? *Journal of Peace Research, 51*(2), 227–243.

Kunkeler, J., & Peters, K. (2011). The boys are coming to town: Youth, armed conflict and urban violence in developing countries. *International Journal of Conflict and Violence, 5*(2), 277–291.

Lavandaio, E. (2008). *Conozcamos más sobre minería*. Buenos Aires: Instituto de Geología y Recursos Minerales, Servicio Geológico Minero Argentino.

Le Billon, P. (2000). *The political economy of war: What relief agencies need to know* Paper no. 33. London: Humanitarian Practice Network, Overseas Development Institute.

Le Billon, P. (2001). The political ecology of war: Natural resources and armed conflicts. *Political Geography, 20*(5), 561–584.

Le Billon, P. (2013). *Wars of plunder: Conflicts, profits and the politics of resources*. Oxford University Press.

Leiteritz, R., Nasi, C., & Rettberg, A. (2009). Para desvincular los recursos naturales del conflicto armado en Colombia. *Colombia Internacional, 70*, 215–229.

Libman, A. (2010). *Subnational resource curse: Do economic or political institutions matter?*, Frankfurt School – working paper series, no. 154. Retrieved from <http://www.frankfurt-school.de/clicnetclm/fileDownload.do?goid=000000248993AB4>.

Lobo, I., & Segura, F. (2010). No todo lo que brilla: múltiples facetas del conflicto en la explotación de oro. In R. Leiteritz, C. Nasi, & A. Rettberg (Eds.), *¿Diferentes recursos, diferentes conflictos? Una exploración de la economía política regional del conflicto armado colombiano*. Bogotá: International Development Research Centre (IDRC), Final Report.

Lujala, P., & Rustad, S. A. (Eds.) (2011). *High-value nautral resources and post-conflict peacebuilding*. Routledge.

Lupsha, P. (1997). La guerra contra las drogas ha muerto. ¡Viva la guerra contra el crimen organizado transnacional. In F. Thoumi, & A. Guáqueta (Eds.), *El rompecabezas de las drogas ilegales en Estados Unidos (Una visión ecléctica)* (pp. 53–86). Bogotá, D.C.: Ediciones Uniandes.

Mähler, A., Shabafrouz, M., & Strüver, G. (2011). *Conflict prevention through natural resource management*, GIGA working papers 158. Retrieved from <http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1763851>.

Maldonado, I., & Rozo, L. (2014). Convergencia de los grupos armados organizados al margen de la ley en la minería aurífera aluvial en la subregión del Bajo Cauca antioqueño. *Revista Criminalidad, 59*(3), 119–138.

Massé, F., & Camargo, J. (2012). *Actores armados ilegales y sector extractivo en Colombia*. Bogotá, D.C., Colombia: CITpax Colombia, Observatorio Internacional.

Mehlum, H., Moene, K., & Torvik, R. (2006). Institutions and the resource curse. *The Economic Journal, 116*, 1–20.

Mejía, D. (2012). *¿El siguiente boom ilegal en Colombia? Minería ilegal y violencia* Foco Económico. Retrieved from <http://focoeconomico.org/2012/05/08/el-siguiente-boom-ilegal-en-colombia-mineria-ilegal-y-violencia/>.

Mejía, D., & Gaviria, A. (2011). *Políticas antidroga en Colombia: éxitos, fracasos y extravíos*. Bogotá, D.C.: Ediciones Uniandes.

Mejía, D., & Restrepo, P. (2011). *The war on illegal drugs in producer and consumer countries: A simple analytical framework* Documentos CEDE No. 2. Bogotá, D.C.: Facultad de Economía, Universidad de los Andes.

Mejía, D., Restrepo, P., & Rozo, S. (2014). *On the effects of enforcement on illegal markets: Evidence from a quasi-experiment in Colombia*. Retrieved from <http://www.sandravrozo.com/uploads/2/9/3/0/29306259/mrr_august_18.pdf>.

Mejía, D., Rettberg, A., Idrobo, N., & Ortiz-Riomalo, J. (2013). *Oro, drogas y criminalidad: Una comparación de regiones colombianas*. Bogotá, D.C.: Manuscript work in progress.

Melo, H. (1974). *El mercado internacional del oro y la explotación del oro en Colombia*. Bogotá, D.C.: CID, Universidad Nacional de Colombia.

Ministerio de Defensa Nacional (MDN) (2010). *Logros de la Política de Consolidación de la Seguridad Democrática – PCSD*. Bogotá, D.C.: Retrieved from <http://www.mindefensa.gov.co/irj/go/km/docs/Mindefensa/Documentos/descargas/estudios%20sectoriales/info_estadistica/Logros%20de%20Politica%20CSD%20Junio%202010.pdf>.

Ministerio de Defensa Nacional (MDN) (2011). *Política Nacional de Defensa de la Libertad Personal 2011–2014*. Bogotá, D.C.: Retrieved from <http://www.mindefensa.gov.co/irj/go/km/docs/Mindefensa/Documentos/descargas/Prensa/Documentos/politica-libpersonal.pdf>.

Ministerio de Defensa Nacional (MDN) (2013). *Logros de la Política Integral de Seguridad y Defensa para la Prosperidad – PISDP*. Bogotá, D.C. Retrieved from <http://www.mindefensa.gov.co/irj/go/km/docs/Mindefensa/Documentos/descargas/estudios%20sectoriales/info_estadistica/Logros_Sector_Defensa.pdf>.

Ministerio de Minas y Energía (MME) (2006). *Colombia país minero. Plan Nacional para el Desarrollo Minero*. Bogotá: Unidad de Planeación Minero Energética – UPME, Retrieved from <http://www.upme.gov.co/Docs/PNDM_2019_Final.pdf>.

Moser, C., & McIlwaine, C. (2001). *Violence in a post-conflict context: Urban poor perceptions from Guatemala*. Washington, DC: The World Bank.

Muggah, R. (Ed.) (2009). *Security and post conflict reconstruction: Dealing with fighters in the aftermath of war*. Routledge.

Nichols, S., Lujala, P., & Bruch, C. (2011). When peacebuilding meets the plan: Natural resource governance and post-conflict recovery. The Whitehead Journal of Diplomacy and International Relations*, 12*.

Nitzschke, H. (2003). *Transforming war economies: Challenges for peacemaking and peacebuilding* New York: Report of the 725th Wilton Park conference, in association with the International Peace Academy Wiston House, Sussex University 27–29. .

O'Callaghan, G. (1993). *The structure and operation of the world gold market*. Washington, DC: International Monetary Fund.

Pardo, R. (2000). Colombia's two-front war. *Foreign Affairs, 79*(4), 64–73.

Paris, R. (2004). *At war's end: Building peace after civil conflict*. New York: Cambridge University Press.

Policía Nacional de Colombia (2013). *Informe integral de gestión institucional año 2013*. Bogotá: Policía Nacional, Oficina de Planeación, Retrieved from <http://www.policia.gov.co/portal/page/portal/INSTITUCION/balance_gestion/Informe_Integral_de_la_Gesti%F3n_Institucional_IV_trimestr.pdf>.

Presidencia de la República (2014). *La minería criminal es un objetivo de alto valor, advirtió el Presidente Santos. Sistema Informativo del Gobierno*. Bogotá: Presidencia de la República, Retrieved from <http://wsp.presidencia.gov.co/Prensa/2014/Enero/Paginas/20140117_04-La-mineria-criminal-objetivo-de-alto-valor-advirtio-el-Presidente-Santos.aspx>.

Puentes, A. (2006). *El opio de los Talibán y la coca de las FARC. Transformaciones de la relación entre actores armados y narcotráfico en Afganistán y Colombia*. Bogotá: Ediciones Uniandes.

Restrepo, J., & Aponte, D. (2009). *Guerra y violencias en Colombia: herramientas e interpretaciones*. Bogotá, D.C.: Editorial Universidad Javeriana.

Rettberg, A. (2010). Global markets, local conflict: Violence in the Colombian coffee region after the ICA breakdown. *Latin American Perspectives, 37*(2), 111–132.

Rettberg, A. (Ed.) (2012). *Construcción de paz en Colombia*. Bogotá D.C: Ediciones Uniandes.

Rettberg, A., Leiteritz, R., & Nasi, C. (2011). Entrepreneurial activity in the context of violent. *Journal of Small Business and Entrepreneurship, 24*(2), 179–196.

Rettberg, A., Norza, E., Ortiz-Riomalo, J., Romero, M., & Romero, P. (2013). *Incertidumbre, vacíos y variación: Un diagnóstico de la actividad minera de oro en diez departamentos de oro* Bogotá, D.C.: Paper delivered at the "Séptimo Congreso Latinoamericano de Ciencia Política" of the Latin American Political Science Association, September 25th–27th, Universidad de los Andes.

Rettberg, A., & Ortiz-Riomalo, J. (2013). *Conflicto dorado: Canales y mecanismos de la relación entre minería de oro, conflicto armado y criminalidad en Colombia*. Centro de Estudios sobre Seguridad y Drogas (CESED), Universidad de los Andes.

Reuter, P., & Trautmann, F. (2009). *A report on global illicit drugs markets 1998–2007*. Netherlands: European Commission, Trimbos Instituut, Rand Europe.

Roache, S., & Rossi, M. (2009). *The effects of economic news on commodity prices: Is gold just another commodity?*. Washington, DC: International Monetary Fund, WP/09/140.

Robinson, J., Torvik, R., & Verdier, T. (2006). Political foundations of the resource curse. *Journal of Development Economics, 79*(2), 447–468.

Rodrik, D. (2007). *One economics, many recipes: Globalization, institutions, and economic growth*. Princeton University Press.

Rodrik, D., Subramanian, A., & Trebbi, F. (2004). Institutions rule: The primacy of institutions over geography and integration in economic development. *Journal of Economic Growth, 9*(2), 131–165.

Rojas, J. H. (2012). *Programa de formalización minera*. Medellín: Octava Rueda de Negocios, September 25th.

Romero, M., Moreno, J., Ladino, J., Sandoval, C., Gutierrez, A., & y Torres, P. (2014). *Minería ilícita, criminalidad, pobreza y vacíos jurídicos en cincuenta y dos municipios en Colombia* Investigación Criminológica, IV (2). Bogotá: Policía Nacional e Imprenta Nacional.

Ross, M. (2004). How do natural resources influence civil war? Evidence from thirteen cases. *International Organization, 58*(1), 35–67.

Ross, M. (2003). Oil, drugs, and diamonds: The varying role of natural resources in civil war. In K. Ballentine, & J. Sherman (Eds.), *Beyond greed and grievance: The political economy of armed conflict*. Boulder: Lynne Rienner Publishers.

Ruiz, M. (2015). *Vidas paralelas* Semana.com. <http://www.semana.com/nacion/articulo/las-vidas-paralelas-de-rafael-colon-pastor-alape/432061-3>.

Rustad, S., & Binningsbø, H. (2012). A price worth fighting for? Natural resources and conflict recurrence. *Journal of Peace Research, 49*(4), 531–546.

Rustad, S., Buhaug, H., Falch, A., & Gates, S. (2011). All conflict is local: Modeling sub-national variation in civil conflict risk. *Conflict Management and Peace Science, 28*(1), 15–40.

Sánchez, F., Ortiz, G., & Moussa, N. (1998). *Panorama minero de América Latina: inversión en la década de los noventa* Serie Medio Ambiente y Desarrollo. Santiago de Chile: Comisión Económica para América Latina y el Caribe (CEPAL).

Sieber, N. (2013). *Health impacts of metal mining & smelting* Bogotá, D.C.: Presentation delivered at the International Conference "Minería en Latinoamérica: retos y oportunidades", Bogota May 23rd–24th, Universidad de los Andes.

Snyder, R. (2006). Does lootable wealth breed disorder? A political economy of extraction framework. *Comparative Political Studies, 39*(8), 943–968.

Tabares, E., & Rosales, R. (2005). *Políticas de control de oferta de coca: "la zanahoria" y "el garrote"* Documentos CEDE no. 10. Bogotá, D.C.: Facultad de Economía, Universidad de los Andes.

The Economist (2013). *Why is less cocaine coming from Colombia?*. Retrieved April 4, 2013, from The Economist web site <http://www.economist.com/blogs/economist-explains/2013/04/economist-explains-why-colombia-produces-less-cocaine>.

Theije, M., & Heemskerk, M. (2009). Moving frontiers in the amazon: Brazilian small-scale gold miners in Suriname. *European Review of Latin American and Caribbean Studies, 87*, 5–25.

Thoumi, F. (1997). Las políticas contra las drogas en Estados Unidos y la certificación de Colombia. In *El rompecabezas de las drogas ilegales en Estados Unidos (una visión ecléctica)* (pp. 211–232). Bogotá, D.C.: Ediciones Uniandes.

Thoumi, F. (2002). *El imperio de la droga. Narcotráfico, economía y sociedad en Los Andes*. Bogotá, D.C.: Ediciones Uniandes.

Tickner, A., & Cepeda, C. (2011). Las drogas ilícitas en la relación Colombia-Estados Unidos: balance y perspectivas. In A. Gaviria, & D. Mejía (Eds.), *Políticas antidroga en Colombia: éxitos, fracasos y extravíos* (pp. 205–228). Bogotá, D.C.: Ediciones Uniandes.

UNODC (2013a). *World drug report*. New York: United Nations.

UNODC (2013b). *Colombia. Monitoreo de cultivos de coca 2012*. Bogotá, D.C.: UNODC, Oficina de las Naciones Unidas contra la Droga y el Delito en Colombia. Gobierno de Colombia.

UPDADH (1989). Una política equivocada: La "guerra" de los Estados Unidos contra la producción de narcóticos en los Andes. *Colombia Internacional, 5*, 36–38.

Uribe, J. D. (2014). *El mercado del oro en Colombia*. Bogotá: Banco de la República, Retrieved from <http://www.banrep.gov.co/sites/default/files/publicaciones/archivos/rbr_nota_1035.pdf>.

WOLA (1993). *¿Peligro inminente? Las FF.AA. de Estados Unidos y la guerra contra las drogas*. Bogotá, D.C.: CEI, IEPRI, Tercer Mundo Editores.

Yassin, N. (2008). *The Thorny road to sustainable peace: The mutation of violence in post-conflict cities* London: Paper presented at the "Cities in an insecure world conference, November 15th, London School of Economics and Political Science and Overseas Development Institute (ODI). Retrieved from <http://www.odi.org.uk/sites/odi.org.uk/files/odi-assets/events-documents/2>.